MIL-E-43058A

20 April 1970
SUPERSEDING
MIL-E-43058 (QMC)
18 January 1962

MILITARY SPECIFICATION

ETCHERS, PNEUMATIC, PORTABLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification covers a portable, pneumatically operated etcher.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard

MILITARY

MIL-P-116 - Preservation, Methods Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5130

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* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL BUREAU OF STANDARDS

Handbook H 28 - Screw-Thread Standards For Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18-67 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street N. W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill. 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Standard product.- The etcher covered by this specification shall be the manufacturer's standard commercial product except for any changes necessary to comply with this specification.

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* 3.2 Material.- Material not definitely specified shall be of the quality normally used by the manufacturer for etchers, providing the completed item complies with all the provisions of this specification.

* 3.3 Construction.- The etcher shall be a portable, hand-held, air operated unit, complete with stylus and air hose. All parts shall be accessible for adjustment, replacement and repair. The etcher shall be provided with either an integral or separate air strainer.

* 3.3.1 Stylus.- The etcher shall be furnished with a stylus in accordance with the manufacturer's commercial practice. The stylus shall have a carbide tip, ground to a 60 to 90 degree included angle. The hardness of the tip shall be Rockwell A 90-92 when tested as specified in 4.3.1.

* 3.3.2 Hose.- The etcher shall be furnished with not less than 8 feet of synthetic rubber air hose in accordance with the manufacturer's current commercial practice.

* 3.3.3 Threaded parts.- All threaded parts shall be in the inch system and shall comply with Handbook H 28.

* 3.3.4 Dimensions and weight.- The etcher shall not exceed 8 inches in overall length, 1-1/4 inch diameter in gripping area, and .1 pound in weight.

3.4 Performance.- The etchers, after a storage period of 72 hours at a temperature of minus 60 to 65 F., shall be capable of marking the surfaces of various types of materials such as hardened steel, aluminum, glass, ceramics and plastics. The etcher shall have adequate gripping surface to permit ease of holding while operating the etcher. The etcher shall be capable of etching a mark of at least 0.002 inch depth in material being marked when tested as specified in 4.3.2.

* 3.4.1 Operation.- The etcher shall operate from an air supply of 100 pounds per square inch gage (psig) maximum, with throttle or regulator adjustment provided to control speed for marking of various material specified in 3.4. The regulator shall be marked to permit easy adjustment and control.

3.4.2 Excessive air pressure.- The etcher and all its components shall operate without damage when tested as specified in 4.3.3.

3.4.3 Continuous duty and shock.- The etcher shall show no signs of damage or loosening of components when tested as specified in 4.3.4 and 4.3.5

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* 3.5 Protective coating.- All bare metal surfaces shall be painted or electro-chemical finished in accordance with the manufacturer's commercial practice.

* 3.6 Extra styli.- Two extra styli shall be furnished with the etcher.

* 3.7 Marking.- The etcher shall be marked with the manufacturer's identification or trademark and model number.

* 3.8 Workmanship.- The etcher shall be free of any slivers, chips, splinters and rough or sharp edges. All screws and other fasteners shall be tight and all components shall be aligned. There shall be no voids in plating or coating, imbedded foreign matter, or blisters on plated or coated surfaces.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials should be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.2.1.1 Intermediate testing.- Testing for hardness of the stylus tip shall be performed in accordance with 4.3.1 on each sample unit prior to assembly. For this test, a lot shall consist of all styli heat treated as a batch or under the same process and temperature control. The sample unit shall be one stylus. The inspection level shall be S-2 with an acceptable quality level (AQL) of 1.5, expressed in terms of defects per hundred units. Any reading not within the specified range shall constitute a defect.

* 4.2.2 End item inspection.- A lot shall consist of all etchers offered for inspection at one time. The sample unit shall be one completely fabricated etcher.

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* 4.2.2.1 Visual examination.- The completely assembled etcher shall be examined for defects in table I. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not plated or coated where required	X	
	Signs of corrosion	X	
	Foreign matter imbedded or blisters		X
	Not as specified	X	
Construction and workmanship	Sharp or rough edges, splinters, slivers, chips	X	
	Screws and other fasteners not tight		X
	Extra stylii missing		X
	Air strainer missing	X	
	Gripping surface not adequate for easy holding	X	
Hose and stylus	Not as specified	X	
Marking for identification	Illegible, incorrect or missing		X

* 4.2.2.2 Dimensional examination.- Inspection shall be made of the etcher for compliance with dimensional and weight requirements. Any deviation from the specific requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

* 4.2.2.3 End item testing.- The initial production unit shall be tested as specified in 4.3.2 through 4.3.5 without evidence of failure. Each sample unit shall be tested as specified in 4.3.2 using an inspection level of S-2 with an AQL of 1.5 defects per hundred units.

* 4.2.2.4 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, packing and marking as

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required by section 5 are complied with. Defects shall be as indicated in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be all of the containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0, expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examine	Defect
Markings, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Preservation	Preservative improperly applied or missing
Materials	Component missing or damaged
Workmanship	Inadequate application of components, such as bulging or distortion of containers
Weight	Weight of contents per shipping container is more than specified

4.2.2.4.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure and strapping shall be in accordance with the appendix of that specification.

4.3 Tests.-

* 4.3.1 Hardness test.- The stylus tip shall be tested for Rockwell hardness in accordance with ASTM E 18-67.

* 4.3.2 Performance test.- The etcher shall be stored at an ambient temperature between minus 65 F. and minus 60 F. for 72 hours. The etcher shall then be stabilized at an ambient temperature between minus 25 F. and minus 20 F. The etcher shall then be connected to a maximum air pressure of 100 psig. Hardened steel, aluminum, glass, ceramics and plastic panels shall be etched. The marking portion of the test shall be repeated at

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ambient temperatures of between plus 70 F. and plus 90 F. and between plus 110 F. and plus 115 F. Failure of the etcher to mark to a depth of 0.002 inch shall constitute failure of this test.

* 4.3.3 Excessive air pressure test.- The etcher shall be subjected to a line pressure of 1-1/2 times the design pressure, except the pressure shall be limited to not less than 100 psig nor more than 150 psig with throttle or regulator wide open for a period of 5 minutes. Nonconformance to the requirements of 3.4.2 shall constitute failure of this test.

* 4.3.4 Continuous duty test.- The etcher shall be connected to a maximum air pressure line of 100 psig. The etcher shall be operated at 15 second intervals until the time intervals total 4 hours. Any nonconformance to the requirements of 3.4.3 shall constitute failure of this test.

* 4.3.5 Shock test.- The etcher, less the stylus, shall be dropped once through its major axis onto a concrete surface from a height of 3 feet to determine compliance with 3.4.3. Any signs of crack, chip, or breakage shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Cleaning.- Cleaning shall be accomplished by process C-1 of MIL-P-116.

5.1.1.2 Drying.- Drying shall be accomplished in accordance with one or more drying procedures listed in MIL-P-116.

5.1.1.3 Preservation application.- Immediately after cleaning and drying, interior surfaces of the air motor shall be coated with type P-9 preservative of MIL-P-116, applied through the lubricating system or by feeding the preservative into the air inlet, while operating the motor, until the preservative appears at the exhaust ports. Exterior unpainted surfaces, including hose fittings, clamps and accessories shall be coated with type P-9 preservative.

5.1.1.4 Unit packaging.- Each hose shall be coiled to the minimum safe diameter and secured with evenly spaced ties of cotton tape. All components shall be unit protected in accordance with Method IA-15 of MIL-P-116.

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5.1.1.5 Intermediate packaging.- Twelve units, packaged as specified in 5.1.1.4, shall be placed in a close fitting intermediate container conforming to the requirements of type CF (variety SW) or SF, class domestic, style RSC of PPP-B-636. Closure shall be in accordance with Method II of the appendix of PPP-B-636.

5.1.2 Level C.- Etchers shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.- Etchers, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to grade V2s, style RSC of PPP-B-636. Closure, waterproofing by means of tape and reinforcing shall be in accordance with the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B.- Etchers, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF (variety SW) or SF, class domestic, grade 275, style RSC of PPP-B-636. Closure shall be in accordance with Method II of the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C.- Etchers, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The portable pneumatic etching tool covered by this specification is intended for etching identification markings on glass, plastic and various types and grades of ferrous and non-ferrous metals.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - GL
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Review activity:

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Preparing activity:

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Project No. 5130-0166

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
ETCHERS, PNEUMATIC, PORTABLE MIL-E-43058A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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